

Spectral Green's function

The free-space Green's function in the spectral domain: derivation

In this document the procedure to derivate the Green's function of free space (homogenous isotropic media) in the spectral domain starting from its spatial representation is explained in details. First of all let us recall the expression of the free space Green's function in the spatial domain:

$$\tilde{G}_{fs}^{ej}(\vec{r}, \vec{r}') = -j \frac{\zeta}{k} \begin{bmatrix} k^2 + \frac{\partial^2}{\partial x^2} & \frac{\partial^2}{\partial x \partial y} & \frac{\partial^2}{\partial x \partial z} \\ \frac{\partial^2}{\partial y \partial x} & k^2 + \frac{\partial^2}{\partial y^2} & \frac{\partial^2}{\partial y \partial z} \\ \frac{\partial^2}{\partial z \partial x} & \frac{\partial^2}{\partial z \partial y} & k^2 + \frac{\partial^2}{\partial z^2} \end{bmatrix} \begin{bmatrix} e^{-jk|r-r'|} \\ 4\pi|r-r'| \end{bmatrix} \quad (1.1)$$

The spectral representation of a function can be obtained calculating the Fourier transform of the domain the function is defined in, which is spatial in this case:

$$G(k_x, k_y) = \int_{-\infty}^{\infty} g(x, y) e^{j(k_x x + k_y y)} dp \quad (1.2)$$

$$g(x, y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(k_x, k_y) e^{-j(k_x x + k_y y)} dk_p \quad (1.3)$$

In order to obtain the spectral expression of the GF the key step is to obtain a spectral expression of the scalar potential ψ , solution of the Helmholtz's equation:

$$\nabla^2 \psi + k^2 \psi = -\delta(\vec{r} - \vec{r}') \quad (1.4)$$

After imposing the radiation condition ψ turns into (which corresponds to the scalar GF in the space domain):

$$\psi = \frac{e^{-jk|r-r'|}}{4\pi|r-r'|} \quad (1.5)$$

We do not know its Fourier transform, but we know we can express ψ via the anti-Fourier transform of its spectral representation $\Psi(\vec{k}, \vec{r}')$ as (note that the anti-Fourier transform is here defined only for $\vec{r}$ and not for $\vec{r}'$, the dependence on $\vec{r}'$ remains in $\Psi(\vec{k}, \vec{r}')$):

$$\psi(\vec{r}, \vec{r}') = \frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} \Psi(\vec{k}, \vec{r}') e^{-jk_x x} e^{-jk_y y} e^{-jk_z z} dk_x dk_y dk_z \quad (1.6)$$

The starting point will be the representation of the delta function in Cartesian coordinates using its inverse Fourier transform. Given the fact that the Fourier transform of a delta distribution has unitary amplitude, and since:

$$FT\{\delta(x - x')\} = \int_{-\infty}^{\infty} \delta(x - x') e^{jk_x x} dx = e^{jk_x x'} \quad (I.7)$$

$$FT^{-1}\{e^{jk_x x'}\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{jk_x x'} e^{-jk_x x} dk_x = \delta(x - x') \quad (I.8)$$

It is easy to verify that:

$$\delta(\vec{r} - \vec{r}') = \frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')} dk_x dk_y dk_z \quad (I.9)$$

From vectorial algebraic operations it is apparent that:

$$\nabla^2 \psi + k^2 \psi = (\nabla^2 + k^2) \psi \quad (I.10)$$

Substituting (I.6),(I.9) into (I.4) we obtain the following expression:

$$\begin{aligned} \frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} (\nabla^2 + k^2) \Psi(\vec{k}, \vec{r}') e^{-jk_x x} e^{-jk_y y} e^{-jk_z z} dk_x dk_y dk_z \\ = -\frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')} dk_x dk_y dk_z \end{aligned} \quad (I.11)$$

Performing the algebraic derivatives introduced by the ∇ operator on the HLS of the equation, since the $\vec{r}$ dependence is present only in the exponentials' argument one arrives to:

$$\begin{aligned} \frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} (k^2 - k_x^2 - k_y^2 - k_z^2) \Psi(\vec{k}, \vec{r}') e^{-jk_x x} e^{-jk_y y} e^{-jk_z z} dk_x dk_y dk_z \\ = -\frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')} dk_x dk_y dk_z \end{aligned} \quad (I.12)$$

This expression is valid for every $\vec{r}$; for this reason it is possible to equate the spectra:

$$(k^2 - k_x^2 - k_y^2 - k_z^2) \Psi(\vec{k}, \vec{r}') e^{-jk_x x} e^{-jk_y y} e^{-jk_z z} = -e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')} \quad (I.13)$$

$$\Psi(\vec{k}, \vec{r}') = -\frac{e^{jk_x x'} e^{jk_y y'} e^{jk_z z'}}{(k^2 - k_x^2 - k_y^2 - k_z^2)} \quad (I.14)$$

Let us now substitute (I.12) Into (I.6) to obtain the spectral representation of the scalar potential, which leads to easy derivatives for the GF:

$$\psi(\vec{r}, \vec{r}') = \frac{e^{-jk|r-r'|}}{4\pi|r-r'|} \quad (I.15)$$

$$= \frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} -\frac{e^{jk_x x'} e^{-jk_y y'} e^{-jk_z z'}}{(k^2 - k_x^2 - k_y^2 - k_z^2)} e^{-jk_x x} e^{-jk_y y} e^{-jk_z z} dk_x dk_y dk_z$$

$$\frac{e^{-jk|r-r'|}}{4\pi|r-r'|} = \frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} -\frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')}}{(k^2 - k_x^2 - k_y^2 - k_z^2)} dk_x dk_y dk_z \quad (I.16)$$

The complex spatial derivatives introduced by the Green's function dyad are now analytical and can be easily performed as:

$$\begin{array}{lll} \frac{\partial^2}{\partial x^2} \psi(\vec{r}, \vec{r}') = -k_x^2 & \frac{\partial^2}{\partial x \partial y} \psi(\vec{r}, \vec{r}') = -k_y k_x & \frac{\partial^2}{\partial x \partial z} \psi(\vec{r}, \vec{r}') = -k_x k_z \\ \frac{\partial^2}{\partial y \partial x} \psi(\vec{r}, \vec{r}') = -k_y k_x & \frac{\partial^2}{\partial y^2} \psi(\vec{r}, \vec{r}') = -k_y^2 & \frac{\partial^2}{\partial y \partial z} \psi(\vec{r}, \vec{r}') = -k_y k_z \\ \frac{\partial^2}{\partial z \partial x} \psi(\vec{r}, \vec{r}') = -k_z k_x & \frac{\partial^2}{\partial z \partial y} \psi(\vec{r}, \vec{r}') = -k_z k_y & \frac{\partial^2}{\partial z^2} \psi(\vec{r}, \vec{r}') = -k_z^2 \end{array}$$

The Green's function can now be re-written as:

$$\tilde{G}_{fs}^{ej}(\vec{r}, \vec{r}') = j \frac{\zeta}{k} \frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} \begin{bmatrix} k^2 - k_x^2 & -k_y k_x & -k_x k_z \\ -k_y k_x & k^2 - k_y^2 & -k_y k_z \\ -k_z k_x & -k_z k_y & k^2 - k_z^2 \end{bmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')}}{(k^2 - k_x^2 - k_y^2 - k_z^2)} dk_x dk_y dk_z \quad (I.17)$$

(I.17) represents the spectral free-space Green's function for the electric current. The most important aspect is that now the spatial derivatives that appeared in the spatial representation are now solved. However, note that the expression is not fully analytical: the triple integral still has to be performed.

IMPORTANT:

$$\psi(\vec{r}, \vec{r}') = \frac{e^{-jk|r-r'|}}{4\pi|r-r'|} = \frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} -\frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')}}{(k^2 - k_x^2 - k_y^2 - k_z^2)} dk_x dk_y dk_z$$

$$\tilde{G}_{fs}^{ej}(\vec{r}, \vec{r}')$$

$$= j \frac{\zeta}{k} \frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} \begin{bmatrix} k^2 - k_x^2 & -k_y k_x & -k_x k_z \\ -k_y k_x & k^2 - k_y^2 & -k_y k_z \\ -k_z k_x & -k_z k_y & k^2 - k_z^2 \end{bmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')}}{(k^2 - k_x^2 - k_y^2 - k_z^2)} dk_x dk_y dk_z$$

2D Integration

Although the complete expression for the GF is derived, (I.17) represents a triple integral; we typically deal with 2D integrals rather than 3D. Thus, we will now derive a double integral representation based on the following mathematical identity:

$$\frac{e^{-jk|r-r'|}}{4\pi|r-r'|} = -j \frac{1}{8\pi^2} \iint_{-\infty}^{\infty} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-j\sqrt{k^2-k_x^2-k_y^2}|z-z'|}}{\sqrt{k^2-k_x^2-k_y^2}} dk_x dk_y \quad (II.1)$$

The derivation of this expression will be given later. The important consequence is that thanks to this identity the free-space Green's function becomes now:

$$\begin{aligned} \tilde{G}_{fs}^{ej}(\vec{r}, \vec{r}') = & -j \frac{\zeta}{k} \frac{1}{8\pi^2} \iint_{-\infty}^{\infty} \begin{bmatrix} k^2 - k_x^2 & -k_y k_x & -k_x(\pm k_z) \\ -k_y k_x & k^2 - k_y^2 & -k_y(\pm k_z) \\ -(\pm k_z)k_x & -(\pm k_z)k_y & k^2 - k_z^2 \end{bmatrix} \\ & \cdot \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-j\sqrt{k^2-k_x^2-k_y^2}|z-z'|}}{\sqrt{k^2-k_x^2-k_y^2}} dk_x dk_y \end{aligned} \quad (II.2)$$

The $\pm k_z$ depends on the observation region: k_z if $z > 0$, $-k_z$ if $z < 0$.

(II.2) is the plane wave representation of electromagnetic fields. The GF is the integral of the superposition of plane waves, due to the fact the dependence from the space variables is only at the exponent. As we already know, plane waves are defined over the entire space. The plane wave expansion represents the total radiated fields as emerging from the source located at (x', y', z') and arriving to the observation point in (x, y, z) . In this scenario, some plane waves have a dominant role over the others.

Now, let us demonstrate (II.1)

Starting from (I.14), we can express it as:

$$\frac{e^{-jk|r-r'|}}{4\pi|r-r'|} = -\frac{1}{(2\pi)^3} \iint_{-\infty}^{\infty} e^{-jk_x(x-x')} e^{-jk_y(y-y')} \left(\int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{(k_\rho^2 - k_z^2)} dk_z \right) dk_x dk_y \quad (II.4)$$

$$k_\rho^2 = k_0^2 - k_x^2 - k_y^2 \quad (II.5)$$

Using the following identity (II.1) is demonstrated:

$$\int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{(k_\rho + k_z)(k_\rho - k_z)} dk_z = j\pi \frac{e^{-jk_\rho|z-z'|}}{k_\rho} \quad (II.6)$$

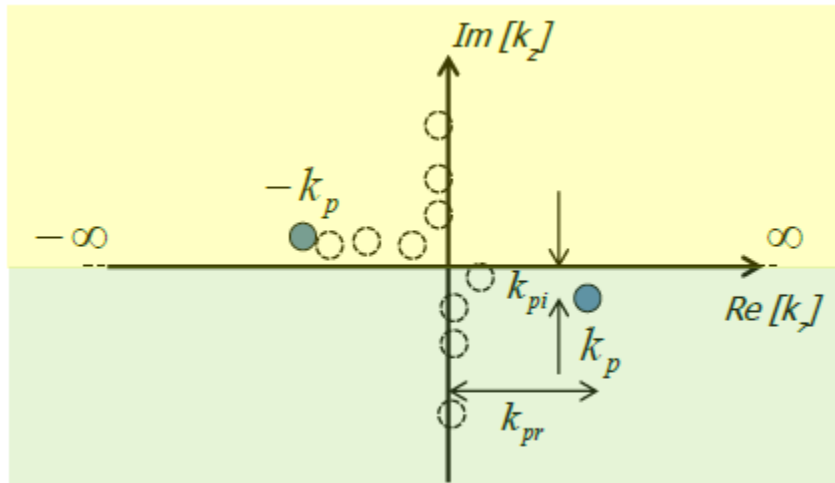
The above identity is based on the complex Cauchy's theorem. Let us go through the steps in details. The integral in (II.6) presents two poles at $k_z = \pm k_\rho$. If we represents k as a complex number, $k = k_r - jk_i$ with $k_r, k_i \geq 0$, we get that:

$$k_\rho = \sqrt{(k_r - jk_i)^2 - k_x^2 - k_y^2} \quad (II.7)$$

Considering $k_x = k_y = 0$:

$$k_\rho = \sqrt{(k_r - jk_i)^2} = k_{\rho r} - jk_{\rho i}, \quad k_{\rho r}, k_{\rho i} \geq 0 \quad (II.8)$$

If the poles of the identity are represented as a function of the real and imaginary part of k_z we can visualize in the k_z complex plane as:



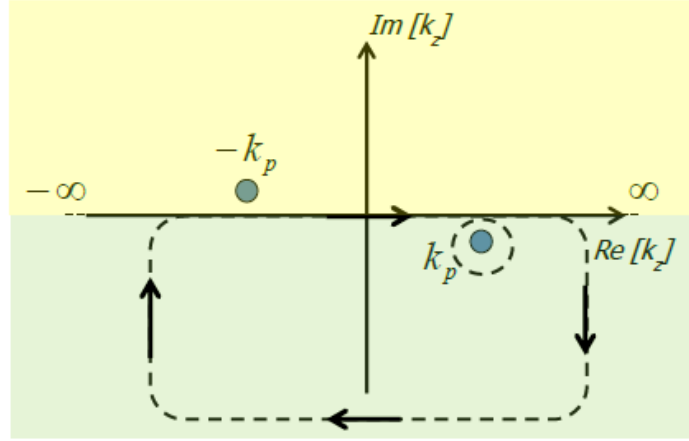
The blue-filled dots represent the location of the poles when $k_x = k_y = 0$. If instead $k_x, k_y \neq 0$ the poles move as the dashed dots.

Now, to close the integral in (II.6) analytically we need to ensure its converges. It is easy to verify that the exponential term $e^{-jk_z(z-z')}$ converges for $k_z \rightarrow \infty$ if:

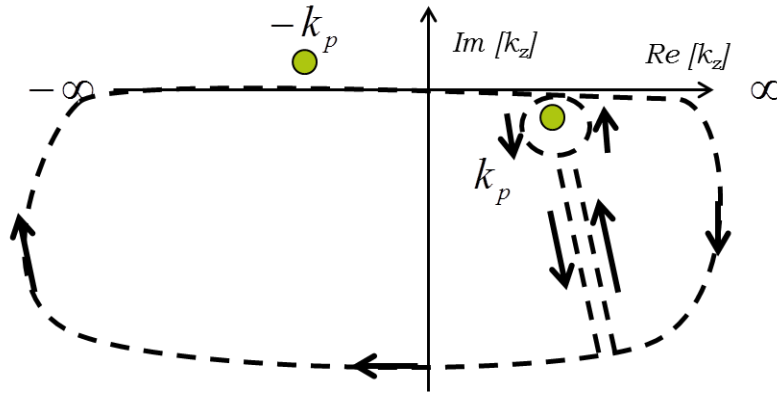
- $(z - z') < 0$ when $\Im\{k_z\} > 0$
- $(z - z') > 0$ when $\Im\{k_z\} < 0$

$\Im\{k_z\} \lessgtr 0$ defines two different half-planes in the k_z complex plane. We will solve the integral for both of them.

I. $\Im\{k_z\} < 0$



First of all the Cauchy's theorem states that if the integration path is closed and not crossing/enclosing any singularity the result of the integral is zero. To avoid the pole while closing the integral we need the following integration path:



The total path will be the sum of three integrations:

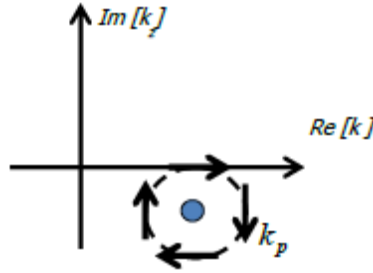
$$I + I_{up} + I_{CCKW} + I_{down} + I_{lhs} = 0 \quad (II.9)$$

I_{CCKW} represents the counter clockwise integration around the singularity, I_{lhs} is the integral performed to the lower half space (lhs) at an infinite distance ($\Im\{k_z\} \ll 0$) where the exponential term forces it to be zero, I_{up} and I_{down} are the integrals going to and coming from the pole which are equal and opposite ($I_{up} = -I_{down}$), while I is the integral that we are trying to solve.

$$\begin{aligned} I &= \int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{(k_\rho + k_z)(k_\rho - k_z)} dk_z \psi(\vec{r}, \vec{r}') \\ &= \frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} \Psi(k, \vec{r}') e^{-jk_x x} e^{-jk_y y} e^{-jk_z z} dk_x dk_y dk_z \end{aligned} \quad (II.10)$$

Since I_{lhs} is zero thanks to the exponential term. The intended contour is closed at infinity by a semicircle of infinite radius in the lhs, which result in an exponential attenuation. The integral reduces to the contribution of the residue associated to the pole with negative imaginary part:

$$I = -I_{CCKW} = I_{CKW} \quad (II.11)$$



Where I_{CKW} stands for clockwise: we inverted the integration path.

We can rewrite the integral as:

$$I = \int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{(k_\rho + k_z)(k_\rho - k_z)} dk_z = \oint_{ClockWise} \frac{F^-(k_z)}{(k_\rho - k_z)} dk_z \quad (II.12)$$

$$F^-(k_z) = \frac{e^{-jk_z(z-z')}}{(k_\rho + k_z)} \quad (II.13)$$

If $F^-(k_z)$ is an analytical function, we can solve the integral in two steps performing a change of variable in each step:

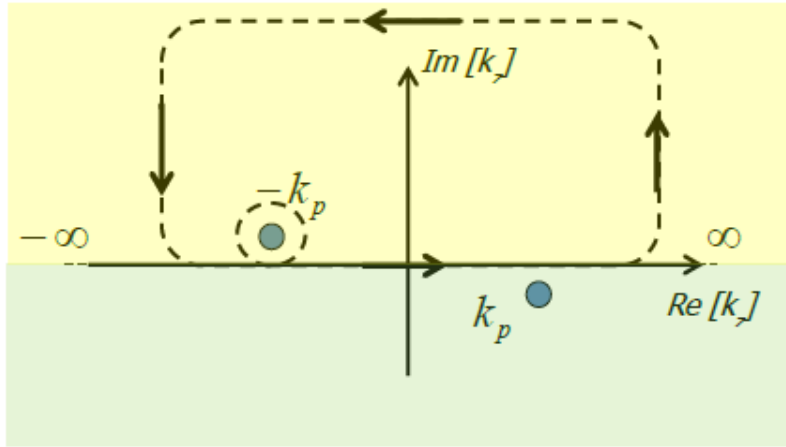
$$\oint_{ClockWise} \frac{F^-(k_z)}{(k_\rho - k_z)} dk_z \quad (II.14)$$

- i. We define $s = (k_\rho - k_z)$, then $I = \int_{ClockWise} \frac{F^-(k_\rho - s)}{s} ds$
- ii. We define $s = \rho e^{-j\phi}$, where ρ is arbitrarily small and the integral to the limit $\rho \rightarrow 0$ becomes:

$$\begin{aligned} I &= -\lim_{\rho \rightarrow 0} \int_0^{2\pi} \frac{F^-(k_\rho - \rho e^{-j\phi})}{\rho e^{-j\phi}} \frac{ds}{d\phi} d\phi = -F^-(k_\rho) \int_0^{2\pi} \frac{1}{\rho e^{-j\phi}} (-j\rho e^{-j\phi}) d\phi \\ &= jF^-(k_\rho) \int_0^{2\pi} d\phi = 2\pi jF^-(k_\rho) \end{aligned} \quad (II.15)$$

II. $\Im\{k_z\} > 0$

The same steps can be applied for the upper half-space (uhs):

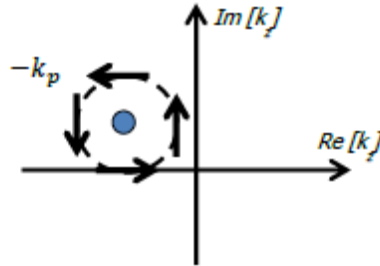


As before:

$$I + I_{down} + I_{CKW} + I_{up} + I_{lhs} = 0 \quad (II.16)$$

Which becomes:

$$I = -I_{CKW} = I_{CCKW} \quad (II.17)$$



Thus, we can rewrite the integral as:

$$I = \int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{(k_p + k_z)(k_p - k_z)} dk_z = \oint_{CounterClockWise} \frac{F^+(k_z)}{(k_p + k_z)} dk_z \quad (II.18)$$

$$F^+(k_z) = \frac{e^{-jk_z(z-z')}}{(k_p - k_z)} \quad (II.19)$$

If $F^+(k_z)$ is an analytical function, we can solve the integral in two steps performing a change of variable in each step:

$$\oint_{CounterClockWise} \frac{F^+(k_z)}{(k_\rho + k_z)} dk_z \quad (II.20)$$

- i. We define $s = (k_\rho + k_z)$, then $I = \int_{CounterClockWise} \frac{F^+(s-k_\rho)}{s} ds$
- ii. We define $s = \rho e^{-j\phi}$, where ρ is arbitrarily small and the integral to the limit $\rho \rightarrow 0$ becomes:

$$\begin{aligned} I &= \lim_{\rho \rightarrow 0} \int_0^{2\pi} \frac{F^+(\rho e^{-j\phi} - k_\rho)}{\rho e^{-j\phi}} \frac{ds}{d\phi} d\phi = F^+(-k_\rho) \int_0^{2\pi} \frac{1}{\rho e^{-j\phi}} (j\rho e^{-j\phi}) d\phi \\ &= jF^+(-k_\rho) \int_0^{2\pi} d\phi = 2\pi jF^+(-k_\rho) \end{aligned} \quad (II.21)$$

Considering now both the solutions for the lhs and uhs we have:

$$F^-(k_z) = \frac{e^{-jk_z(z-z')}}{(k_\rho + k_z)}; \quad F^+(k_z) = \frac{e^{-jk_z(z-z')}}{(k_\rho - k_z)} \quad (II.22)$$

Therefore, to ensure the integrand function convergence:

- If $(z - z') > 0$, $I^- = 2\pi jF^-(k_\rho) = 2\pi j \frac{e^{-jk_\rho(z-z')}}{(2k_\rho)} = \pi j \frac{e^{-jk_\rho(z-z')}}{k_\rho}$
- If $(z - z') < 0$, $I^- = 2\pi jF^+(-k_\rho) = 2\pi j \frac{e^{-jk_\rho(z-z')}}{(2k_\rho)} = \pi j \frac{e^{-jk_\rho(z-z')}}{k_\rho}$

If we joint both cases by using the absolute value $|z - z'|$, we obtain:

$$I^\pm = \pi j \frac{e^{-jk_\rho|z-z'|}}{k_\rho} \quad \forall (z - z') \quad (II.23)$$

And the initial identity is demonstrated

$$\int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{(k_\rho + k_z)(k_\rho - k_z)} dk_z = j\pi \frac{e^{-jk_\rho|z-z'|}}{k_\rho} \quad (II.24)$$

$$\frac{e^{-jk|r-r'|}}{4\pi|r-r'|} = -j \frac{1}{8\pi^2} \iint_{-\infty}^{\infty} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-j\sqrt{k^2 - k_x^2 - k_y^2}|z-z'|}}{\sqrt{k^2 - k_x^2 - k_y^2}} dk_x dk_y \quad (II.25)$$

The same derivation is possible for generalized analytical functions $C(k_z)$:

$$\int_{-\infty}^{\infty} C(k_z) \frac{e^{-jk_z(z-z')}}{(k_\rho + k_z)(k_\rho - k_z)} dk_z = j\pi \frac{e^{-jk_\rho|z-z'|}}{k_\rho} C(\pm k_z) \quad (II.26)$$

IMPORTANT:

$$\tilde{G}_{fs}^{ej}(\vec{r}, \vec{r}') = -j \frac{\zeta}{k} \frac{1}{8\pi^2} \iint_{-\infty}^{\infty} \begin{bmatrix} k^2 - k_x^2 & -k_y k_x & -k_x(\pm k_z) \\ -k_y k_x & k^2 - k_y^2 & -k_y(\pm k_z) \\ -(\pm k_z)k_x & -(\pm k_z)k_y & k^2 - k_z^2 \end{bmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-j\sqrt{k^2 - k_x^2 - k_y^2}|z-z'|}}{\sqrt{k^2 - k_x^2 - k_y^2}} dk_x dk_y$$

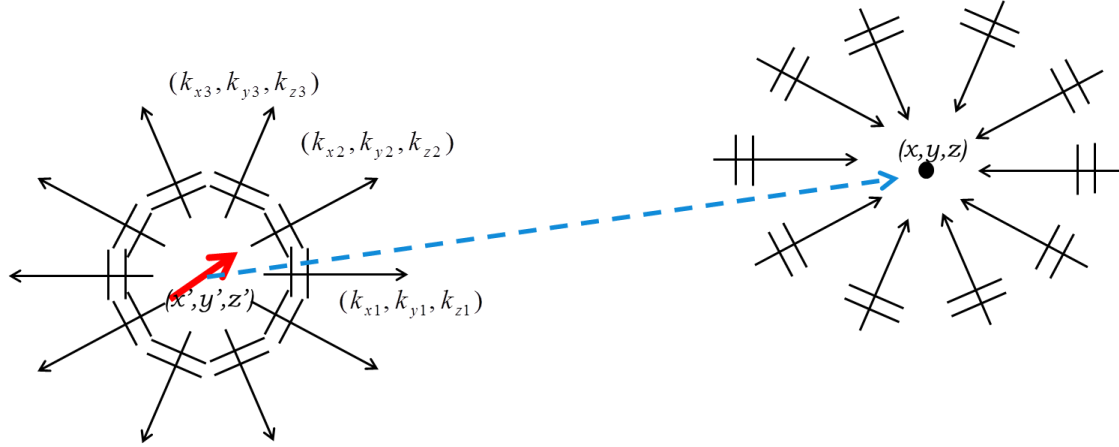
$$\tilde{G}_{fs}^{em}(\vec{r}, \vec{r}') = -\frac{j}{8\pi^2} \iint_{-\infty}^{\infty} \begin{bmatrix} 0 & \pm j k_z & -j k_y \\ \mp j k_z & 0 & j k_x \\ j k_y & -j k_x & 0 \end{bmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-j\sqrt{k^2 - k_x^2 - k_y^2}|z-z'|}}{\sqrt{k^2 - k_x^2 - k_y^2}} dk_x dk_y$$

$$\tilde{G}_{fs}^{hm}(\vec{r}, \vec{r}') = -j \frac{1}{\zeta k} \frac{1}{8\pi^2} \iint_{-\infty}^{\infty} \begin{bmatrix} k_0^2 - k_x^2 & -k_y k_x & -k_x(\pm k_z) \\ -k_y k_x & k^2 - k_y^2 & -k_y(\pm k_z) \\ -(\pm k_z)k_x & -(\pm k_z)k_y & k_0^2 - k_z^2 \end{bmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-j\sqrt{k^2 - k_x^2 - k_y^2}|z-z'|}}{\sqrt{k^2 - k_x^2 - k_y^2}} dk_x dk_y$$

$$\tilde{G}_{fs}^{hj}(\vec{r}, \vec{r}') = \frac{j}{8\pi^2} \iint_{-\infty}^{\infty} \begin{bmatrix} 0 & \pm j k_z & -j k_y \\ \mp j k_z & 0 & j k_x \\ j k_y & -j k_x & 0 \end{bmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-j\sqrt{k^2 - k_x^2 - k_y^2}|z-z'|}}{\sqrt{k^2 - k_x^2 - k_y^2}} dk_x dk_y$$

Far Field asymptotical evaluation: the stationary phase point

The given Green's function represents the plane wave representation of the total field radiated from the source located in (x', y', z') and arriving at the observation point (x, y, z) :



In the far field, the plane wave given by the direct ray from (x', y', z') to (x, y, z) will play a dominant role: it corresponds to a stationary phase point of the integrand function. Thus, mathematically, the far field $\vec{f}^{\text{far}}(\vec{r})$ given by the source $\vec{c}(\vec{r}')$ is:

$$\vec{f}^{\text{far}}(\vec{r}) = \frac{1}{4\pi^2} \iint_{-\infty}^{\infty} \tilde{D}^{fc}(k_x, k_y) \vec{C}(k_x, k_y) e^{-jk_x x} e^{-jk_y y} \frac{e^{-jk_z |z-z'|}}{k_z} dk_x dk_y \quad (\text{III.1})$$

$$\tilde{G}^{fc}(k_x, k_y) = \tilde{D}^{fc}(k_x, k_y) \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z |z-z'|}}{k_z} \quad (\text{III.2})$$

Where $\tilde{D}^{fc}(k_x, k_y)$ is one of the four 3x3 dyads defined at the 'important' section of the previous chapter at pag.10, depending on the source/field combination we are looking for. In (III.1) we can identify in the integrand a function that is varying much slower than the rest around a stationary phase point. We will call this point (k_{xs}, k_{ys}, k_{zs}) , and this function is:

$$A_{\text{slow}}(k_{xs}, k_{ys}) = \tilde{D}^{fc}(k_{xs}, k_{ys}) \vec{C}(k_{xs}, k_{ys}) \quad (\text{III.3})$$

And can be extracted from the integral since this one is dominated by the oscillating term:

$$\vec{f}^{\text{far}}(\vec{r}) = \frac{1}{4\pi^2} \tilde{D}^{fc}(k_{xs}, k_{ys}) \vec{C}(k_{xs}, k_{ys}) \iint_{-\infty}^{\infty} e^{-jk_x x} e^{-jk_y y} \frac{e^{-jk_z |z-z'|}}{k_z} dk_x dk_y \quad (\text{III.4})$$

This means that the product between the Green's function's dyad and the Fourier transform of the source are oscillating much slower than the resto of the integrand. The solution of this integration is now known analytically and is:

$$\begin{aligned}\vec{f}^{far}(\vec{r}) &= j k_z \tilde{G}^{fc}(k_{xs}, k_{ys}, z, z') \vec{C}(k_{xs}, k_{ys}) \frac{e^{-jk_r}}{2\pi r} \\ &= j D^{fc}(k_{xs}, k_{ys}, z, z') \vec{C}(k_{xs}, k_{ys}) \frac{e^{-jk_r}}{2\pi r}\end{aligned}\quad (III.5)$$

(III.5) represents the stationary phase point asymptotical evaluation of the far field.

Now, let us go back to the starting integral (III.1) :

$$\vec{f}^{far}(\vec{r}) = \frac{1}{4\pi^2} \iint_{-\infty}^{\infty} \tilde{D}^{fc}(k_x, k_y) \vec{C}(k_x, k_y) e^{-jk_x x} e^{-jk_y y} \frac{e^{-jk_z |z-z'|}}{k_z} dk_x dk_y \quad (III.6)$$

How do we identify the stationary phase point? First, let us apply a change of variables and switch to the cylindrical integration domain:

$$k_x = k_0 \sin\beta \cos\alpha; \quad k_y = k_0 \sin\beta \sin\alpha; \quad k_z = \sqrt{k_0^2 - k_x^2 - k_y^2} = \pm k_0 \cos\beta \quad (III.7)$$

$$x = r \sin\theta \cos\phi; \quad y = r \sin\theta \sin\phi; \quad z = r \cos\theta \quad (III.8)$$

So that the phase term of the integral becomes:

$$e^{-j\vec{k}\vec{r}} = e^{-jk_0 r (\sin\beta \cos\alpha \sin\theta \cos\phi + \sin\beta \sin\alpha \sin\theta \sin\phi + \cos\beta |\cos\theta|)} \quad (III.9)$$

We need to find the first derivative with respect to the new integrating variables (β, α) of the phase function, which is now:

$$f(\beta, \alpha) = k_0 r (\sin\beta \sin\theta \cos(\alpha - \phi) + \cos\beta |\cos\theta|) \quad (III.10)$$

Pay attention to the fact that the stationary phase point in general depends on the observation half space $(z - z') \lessgtr 0$:

$$\frac{\partial}{\partial \alpha, \partial \beta} f(\beta, \alpha) \quad (III.11)$$

Let us calculate first the derivative of the phase function with respect to α , given the fact that it is not dependent on the observation space:

$$\frac{\partial}{\partial \alpha} f(\beta, \alpha) = -k_0 r (\sin\beta \sin\theta \sin(\alpha - \phi)) = 0 \quad (III.12)$$

This gives us the stationary phase point $\alpha_s = N\pi + \phi$; $N \in \mathbb{N}$. The saddle points are periodical, but the result they represent is not always physical. Indeed, we have to describe the propagation of waves, and we're doing it using a spherical coordinates system. This system is defined for $\theta = [0 \rightarrow \pi], \phi = [0 \rightarrow 2\pi]$. This means that also the saddle points we're going to evaluate the field

at will have to respect these boundaries. Also, for every observation point we perform an integration in (β, α) , which are defined in a spherical coordinate system as well. Moreover, they have to satisfy the domain where the integrand function is defined, that is $(\beta = [0 \rightarrow \pi/2], \alpha = [0 \rightarrow 2\pi])$. For this reason the only possible solution is for $N = 0$: $\alpha_s = \phi$.

Let us calculate the stationary phase point β_s :

- Upper half space observation $(z - z') > 0 \rightarrow \theta < \frac{\pi}{2}, k_z = k_0 \cos \beta$

In this space $|\cos \theta| = \cos \theta$:

$$\frac{\partial}{\partial \beta} f(\beta, \alpha) = k_0 r (\sin \beta \sin \theta \cos(\alpha_s - \phi) + \cos \beta |\cos \theta|) \quad (\text{III.13})$$

$$\begin{aligned} &= k_0 r (\sin \beta \sin \theta + \cos \beta |\cos \theta|) = 0 \\ \sin \beta \sin \theta + \cos \beta |\cos \theta| &= \sin \beta \sin \theta + \cos \beta \cos \theta = \sin(\beta - \theta) = 0 \end{aligned} \quad (\text{III.14})$$

The stationary phase point is $\beta_s = N\pi + \theta$. Since we must satisfy the integration boundaries for $\beta = 0 \rightarrow \pi/2$ the only possible choice is

$$\beta_s = \theta \quad (\text{III.15})$$

- Lower half space observation $(z - z') < 0 \rightarrow \frac{\pi}{2} < \theta < \pi$

$$\frac{\partial}{\partial \beta} f(\beta, \alpha) = k_0 r (\sin \beta \sin \theta \cos(\alpha_s - \phi) + \cos \beta |\cos \theta|) \quad (\text{III.16})$$

$$\begin{aligned} &= k_0 r (\sin \beta \sin \theta + \cos \beta |\cos \theta|) = 0 \\ \sin \beta \sin \theta + \cos \beta |\cos \theta| &= \sin \beta \sin \theta - \cos \beta \cos \theta = \sin(\beta + \theta) \end{aligned} \quad (\text{III.17})$$

The stationary phase point is $\beta_s = N\pi - \theta$. Since we must satisfy the integration boundaries for $\beta = 0 \rightarrow \pi/2$ the only possible choice is

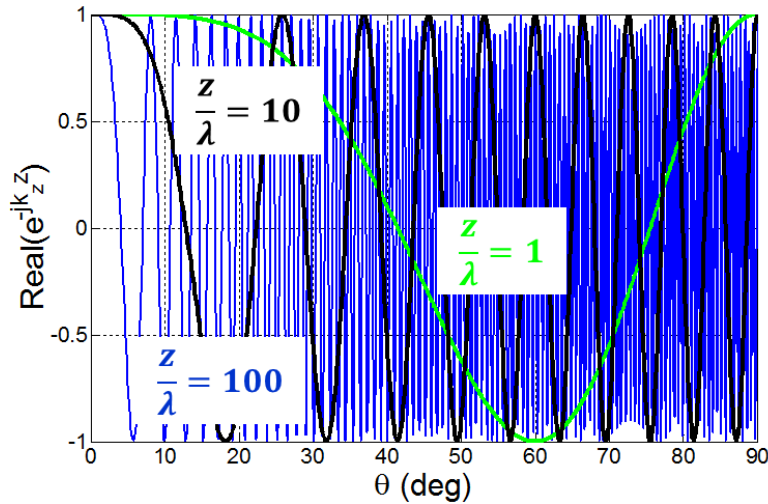
$$\beta_s = \pi - \theta \quad (\text{III.18})$$

Appendix: The Stationary Phase Method

Let us now demonstrate the key steps that led us to this expression. The original integration domain of (III.1) is infinite ($k_x, k_y = (-\infty; \infty)$); when evaluating the far field radiation we can assume that the distance between the source and the observation point $|\vec{r} - \vec{r}'|$ is much larger than the wavelength. For the sake of clarity let us consider a distance observation point in z ; the exponential term is:

$$e^{-jk_z z} = e^{-j\frac{2\pi}{\lambda} \cos\theta z} = e^{-j\left[\frac{2\pi}{\lambda} z\right] \cos\theta} \quad (\text{A.1})$$

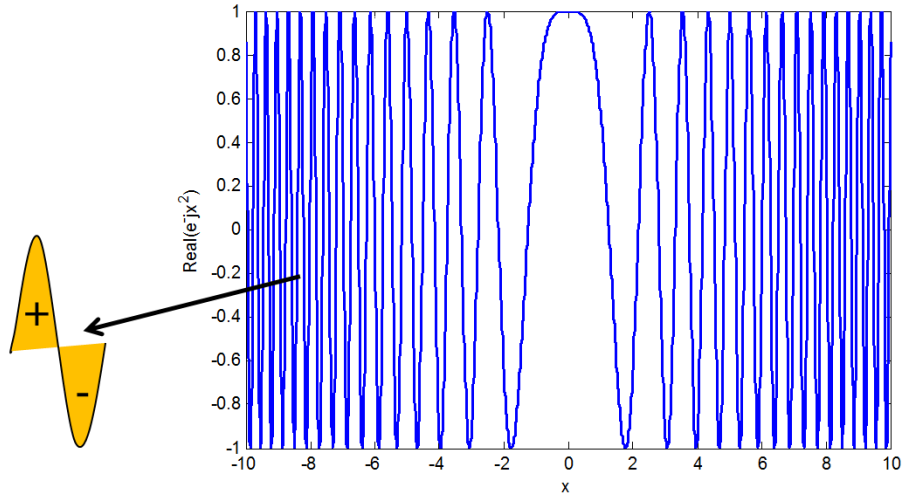
If the constant $\frac{2\pi}{\lambda} z \gg 1$ the exponential term is a highly oscillating function of θ . The higher it is the faster the oscillation:



This means that the contributes will tend to cancel each other out, rendering the calculation of the integral possible with the stationary phase method. To do so, we will use the following identity:

$$I = \int_{-\infty}^{\infty} e^{-j\Omega x^2} = \sqrt{\frac{\pi}{|\Omega|}} e^{\mp j\frac{\pi}{4}} \quad (\text{A.2})$$

The integral of this function is dominated by the area of the portion of the function close to the origin. Indeed, the portion far from the origin (*i.e.* the stationary phase point) oscillates rapidly and leads to destructive contributions

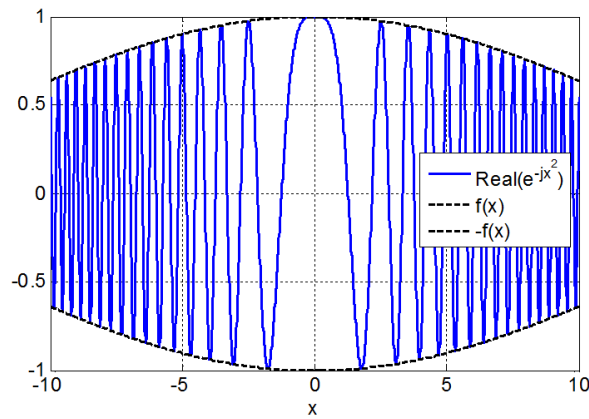


This is given by:

$$\int_{-\infty}^{\infty} e^{-\Omega x^2} = \sqrt{\frac{\pi}{|\Omega|}} \quad (\text{A.3})$$

In (III.1) we can identify exponential terms, which depends on an observation point much greater than the wavelength, multiplied by a function that we said is slowly varying. We can describe this as:

$$I = \int_{-\infty}^{\infty} f(x) e^{-j\Omega x^2} dx \quad (\text{A.4})$$



$f(x)$ is a function that changes negligibly over a cycle of variation $\cos(\Omega x^2)$. To verify this we apply the Taylor's expansion of this function around the origin:

$$f(x) \simeq f(0) + xf'(0) + \frac{x^2}{2} f''(0) + \dots \quad (\text{A.5})$$

It is apparent that the odd-derivatives are zero in the evaluation point. Let us substitute the expansion in the integral:

$$I \simeq f(0) \int_{-\infty}^{\infty} e^{-j\Omega x^2} dx + f''(0) \frac{1}{2} \int_{-\infty}^{\infty} x^2 e^{-j\Omega x^2} dx + f''''(0) \frac{1}{4!} \int_{-\infty}^{\infty} x^4 e^{-j\Omega x^2} dx + \dots \quad (\text{A.6})$$

Using the following identity:

$$\int_{-\infty}^{\infty} x^2 e^{-j\Omega x^2} dx = j \frac{\partial}{\partial \Omega} \int_{-\infty}^{\infty} e^{-j\Omega x^2} dx \quad (\text{A.7})$$

One obtains:

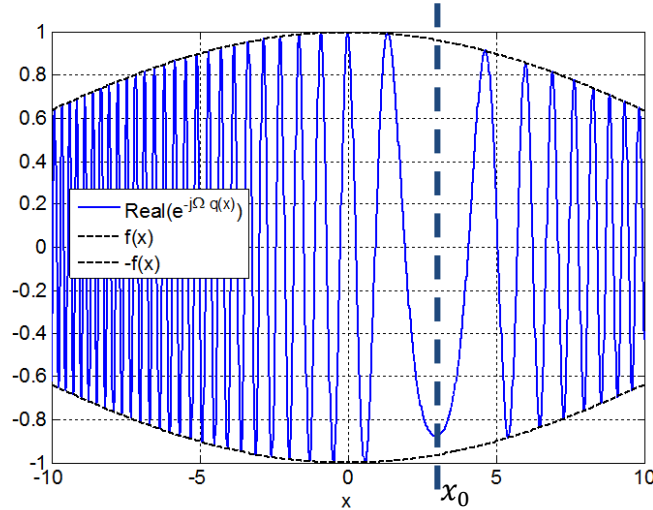
$$I \simeq f(0) \sqrt{\frac{\pi}{j\Omega}} + \frac{1}{2} f''(0) \frac{1}{2} \frac{1}{j\Omega} \sqrt{\frac{\pi}{j\Omega}} + \frac{1}{4!} f''''(0) j \frac{3}{4 - \Omega^2} \sqrt{\frac{\pi}{j\Omega}} + \dots \quad (\text{A.8})$$

Since we are considering observation in the far field $\Omega \gg 1$, and only the first term is significant:

$$I \simeq f(0) \sqrt{\frac{\pi}{j\Omega}} \quad (\text{A.9})$$

If the exponentials have a different x dependence as:

$$I = \int_{-\infty}^{\infty} f(x) e^{-j\Omega q(x)} dx \quad (\text{A.10})$$



Where $q(x)$ is a function with a minimum/maximum in $x_0 \rightarrow q'(x_0) = 0$. The same consideration can be taken now. Let us apply the Taylor's expansion around the stationary phase point x_0 :

$$q(x) \simeq q(x_0) + \frac{1}{2} q''(x_0) (x - x_0)^2 + \dots \quad (\text{A.11})$$

The integral is now dominated by the area around x_0

$$\begin{aligned}
 I &= \int_{-\infty}^{\infty} f(x) e^{-j\Omega q(x_0)} e^{-j\frac{\Omega}{2} q''(x_0)(x-x_0)^2} dx \simeq f(x_0) e^{-j\Omega q(x_0)} \int_{-\infty}^{\infty} e^{-j\frac{\Omega}{2} q''(x_0)(x-x_0)^2} dx \quad (\text{A.12}) \\
 &= f(x_0) e^{-j\Omega q(x_0)} \int_{-\infty}^{\infty} e^{-j\frac{\Omega}{2} q''(x_0)(x')^2} dx'
 \end{aligned}$$